Macropad Build Guide

Oct 6, 2026 · @Antony

You will build a tilted macropad with 12 keys, 2 knobs, 3 sliders and 5 small screens, from loose parts and a 3D-printed case, with no circuit board. It plugs in over USB-C. Plan on two weekends. Every step below has a picture, and none of it needs experience.

What you're building

A 30° wedge with keys, knobs and sliders on top, a small screen in front of each slider and knob, and a slot across the back that holds your phone. Inside is a small computer board called a BlackPill, plus the parts and wires that join them. Everything is joined by hand with solder, so you don't need a circuit board.



Seen from the top, with the phone slot away from you:

	Left						Right
Back row	Slider 1	Slider 2	Slider 3	K1	K2	K3	Knob 2 (E2)
Row 2	↕	↕	↕	K4	K5	K6	Screen S5

	Left						Right
Row 3	↕	↕	↕	K7	K8	K9	Knob 1 (E1)
Front row	↕	↕	↕	K10	K11	K12	
Front edge	S1	S2	S3				Screen S4

The knobs are called **encoders**: they click as you turn them and press like a button. The sliders run the length of the slope. Each screen shows what its slider or knob does. What every key, knob and slider does is set in the firmware, which is still being written; you can change all of it later.

Shopping list and tools

Change each part's status as you order it and it arrives. Parts marked **Have** are already in your box.

Parts

Part	How many	What to look for	Status
STM32F411 "BlackPill" board	1, buy 2	USB-C; the chip must read STM32F411CEU6 and the silver crystal 25.000	To buy
MX key switches	12 (you have 10)		Have
EC11 rotary encoder, 5 pins	2	check it came with a nut	Have
Bourns PTA4543-2015DPB103 slide potentiometer	3	45 mm travel, 10k, **B** = linear	To buy
0.96" OLED screen, 128 x 64, SSD1306	5, buy 6	the **4-pin I2C version (GND VDD SCK SDA), all at address 0x3C**	Have
TCA9548A I2C multiplexer board	1	gives each screen its own channel, since all 5 share one address; sits in the lid	To buy
1N4148 diodes	14, buy 20	small red glass tube with a black band at one end	To buy
100 nF ceramic capacitors	3	one per slider, steadies the reading	To buy

Part	How many	What to look for	Status
6 x 6 mm push buttons, 5 mm tall	2	B (BOOT0) and R (reset), in the lid	Have
Bare tinned copper wire, 0.5 to 0.6 mm (24 to 22 AWG)	about 3 m	the buses: rows, columns, GND and 3V3	To buy
Heat-shrink or PTFE sleeve, 1 to 1.5 mm	about 30 pieces of 8 mm	one wherever a row bus crosses a column bus, so the bare copper can't touch	To buy
Silicone wire, 28 to 30 AWG	several colours	soft wire for the jumpers and the 27 numbered wires to the lid	To buy
M3 x 8 mm countersunk screws	10 (you have 8)	flat heads; 4 for the plate (2 into the frame, 2 into the phone dock), 2 for the dock's strip, 4 for the lid. They cut their own thread in the printed posts, no inserts needed. M3 x 10 grips better	Have
Keycaps, 1u size	12	any MX-style keycaps	To buy
Rubber feet	4		To buy
ST-Link V2 programmer	1, optional	rescues the BlackPill if loading over USB fails; plus 4 male-to-female jumper wires	To buy
Male header strip, 2.54 mm, 40 pins	2	snap into strips for the BlackPill's two rows and the multiplexer, long pins up	To buy
Dupont housings and female crimp pins	1 kit: 1x10 (2), 1x5, 1x3, 1x1	plugs A to E on the header pins (Step 8); a crimp tool helps	To buy
Rainbow ribbon cable, 1.27 mm pitch	about 1 m	the 27 wires from the plate to the lid, as five flat ribbons	To buy

The 2 knobs, the 3 slider caps and the 2 encoder spacer rings are printed (Step 1), so you don't need to buy them.

Tools

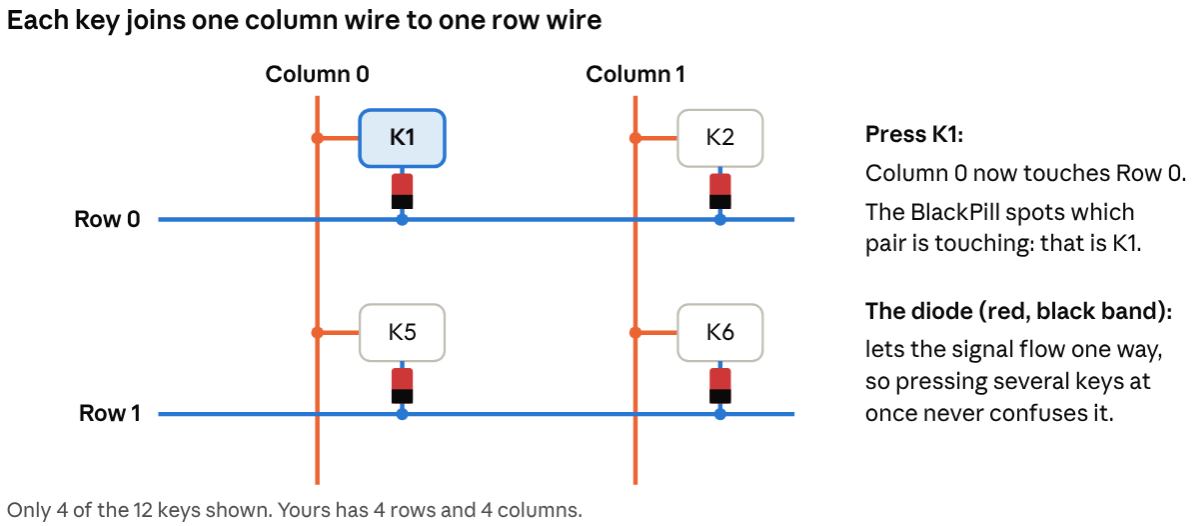
Tool	Why you need it
Soldering iron with temperature control	Joins the wires. Set it to about 330 °C.
Thin solder, 0.8 mm, with flux inside (“rosin core”)	The metal that makes the joints.
Wire strippers	Take the plastic off the wire ends.
Flush cutters	Snip wires and diode legs.
Multimeter with a diode setting	Checks your work before you plug it in.
Tweezers and a helping-hands clamp	Hold small parts while you solder.
3D printer, or a print service	Makes the case.
Hot glue gun	Holds the sliders, screens, BlackPill and buttons in place.
Masking tape and a pen	Label the loose wire ends as you go.
Small screwdriver, a USB-C cable that carries data, a computer	Closing the case and loading the software.

Words you'll meet

Word	What it means
Switch	The part under each key. Pressing it joins its two metal pins.
Encoder	A knob that clicks as you turn it. It also has a push button inside.
Slider	A potentiometer you push along a slot. The board reads where it is, from 0 to 100%.
Screen	A 0.96″ OLED module, 128 x 64 dots, with 4 pins along one edge: GND, VDD, SCK, SDA.
Multiplexer	A small board (TCA9548A) with 8 channels. All 5 screens answer to the same name, so each gets its own channel and the BlackPill picks which one it talks to.

Word	What it means
BlackPill	The small board that runs the macropad. Its holes along the edges are called **pins**, named A4, B13, G, 3V3 and so on.
Diode	A tiny one-way valve for electricity: a red glass body with a black band at one end. The band end is called the cathode.
Capacitor	A tiny part that steadies a signal. The 100 nF ones here have no + or −.
Row and column	The wires that link the keys. Each key sits where one row crosses one column.
Bus	A bare copper wire that many parts solder onto: one per row, one per column, one GND and one 3V3.
Harness	The 27 numbered wires between the plate and the lid, crossing at the hinge.
Solder	Soft metal you melt with the iron to join two wires.
Joint	One spot where you soldered.
G (ground)	The shared "minus" wire everything returns to.
3V3	The 3.3 volt supply from the BlackPill that powers the sliders and screens.
Firmware	The program on the BlackPill that turns key presses into shortcuts.

The keys are wired in a grid. This saves wires: 16 positions (the 12 keys, the 2 knob pushes and 2 spare) need only 8 wires to the BlackPill instead of 15.



how a key press is detected · 4 of the 12 keys

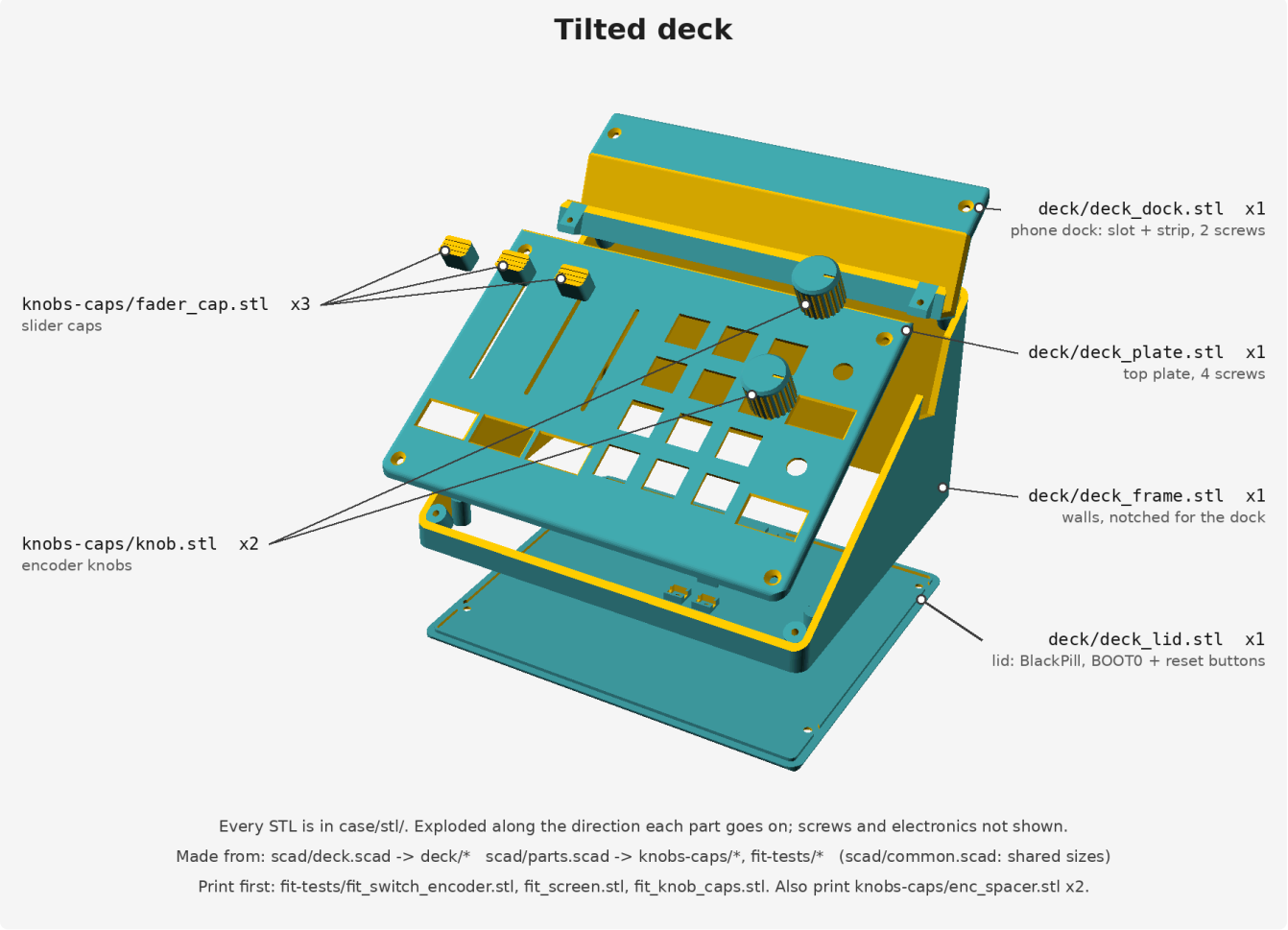
When you press a key, its column wire and row wire touch through the switch. The BlackPill keeps checking every row and column pair, so it knows exactly which key you pressed.

Step 1: Print the test piece, then the case

Everything to print is in `macropad/case/stl/`. Print the small test piece first, so a wrong hole size costs you 15 minutes instead of a whole case.

1. Print `fit-tests/fit_switch_encoder.stl` and `fit-tests/fit_screen.stl`. PLA or PETG works, 0.2 mm layers, 3 walls, 20% infill, no supports.
2. Push one key switch into the square hole. It should **click** in and not fall out when you turn the piece over.
3. Push one encoder's threaded part up through the round hole. It should pass through without forcing.
4. Lay one screen glass-down into the screen test piece. The glass should drop into the pocket, the 4 small pegs should go through its corner holes, and the lit area should fill the window.
5. If any fit is wrong, open `case/scad/common.scad`, change the number, and run `case/build.sh` to make new files:
   - Switch too tight: raise `switch_cut` from 14.0 to 14.1. Too loose: lower it to 13.9.
   - Encoder hole too tight: raise `enc_hole` (now 7.8; 7.3 was too small on the first test print).
   - Screen doesn't fit: measure it and change the `scr_` sizes; pegs too tight for its holes: lower `peg_d`.

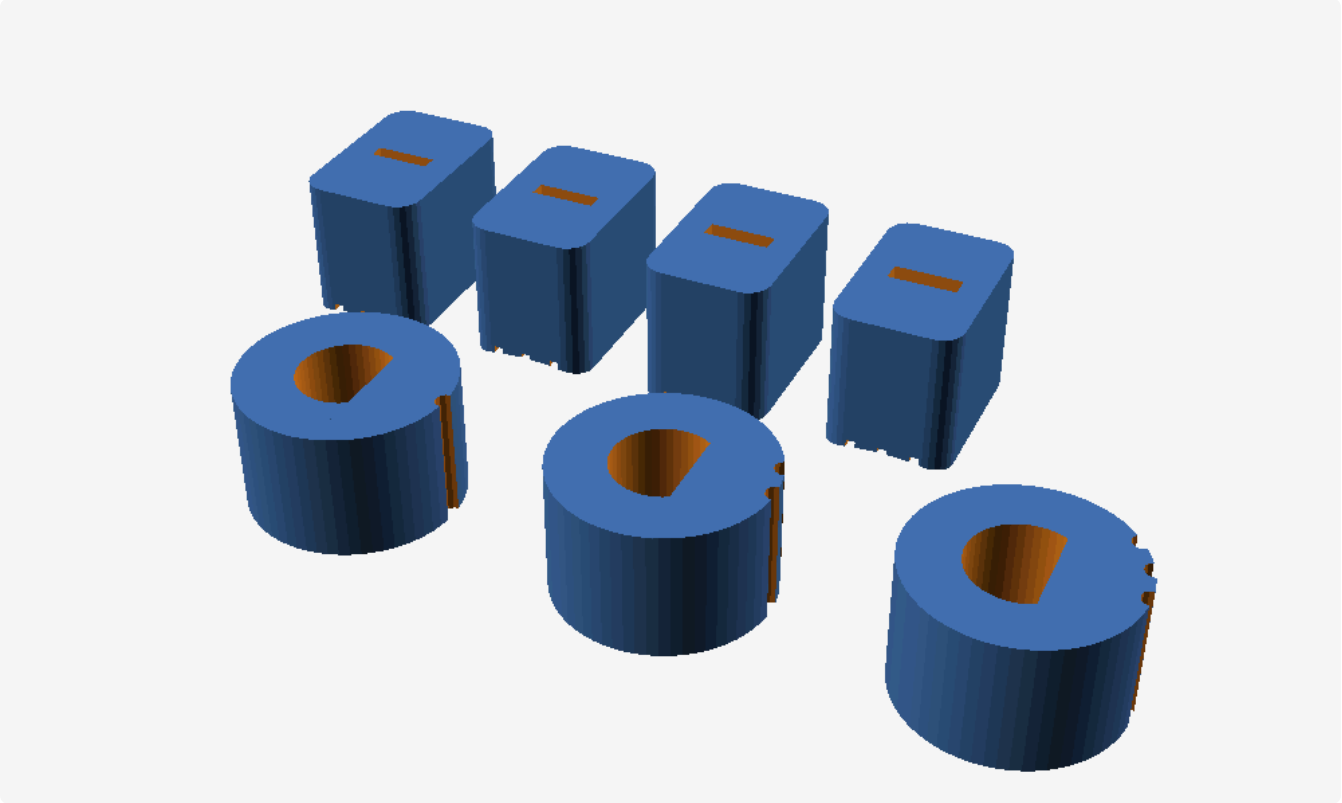
6. Print the case from `deck/`: `deck_plate.stl` (flat, top face down, one piece), `deck_dock.stl` (the phone dock, standing on its slot floor, no supports), `deck_frame.stl` (upright, no supports) and `deck_lid.stl` (flat). All four are already turned the right way up for printing. Every outside edge is rounded, so nothing needs sanding.



Print the knobs. The knob's hole is copied from a Stream Deck+ knob that fits your encoders: 6.16 mm across, with one flat side (a "D"). Measure the shaft from the bottom of the threaded part to the tip: about 15 mm or about 20 mm. Don't cut it; the knob is made just tall enough for it.

1. If your shaft is not 20 mm, open `case/scad/parts.scad`, set `enc_shaft_l` (15 or 20), then run `case/build.sh`. If the flat side is missing, also set `enc_shaft` to `"round"`.
2. Print `fit-tests/fit_knob_caps.stl`. The 3 knobs have holes 6.06, 6.16 and 6.26 mm wide (1, 2 and 3 dots on top). The 4 small blocks are slider caps; they're used below.
3. Push each test knob onto an encoder shaft. Keep the one that goes on firmly by hand and doesn't slip when you turn it, and set `shaft_bore` to its size (6.16 should be right). If one of them is already right, you can simply use it as a real knob.

4. Print `knobs-caps/knob.stl` twice, upside down as it comes, no supports. Also print `knobs-caps/enc_spacer.stl` twice: 3 mm rings that make the encoders sit lower (Step 3).



Print the slider caps and cut the screw threads. Push the 4 test caps from `fit_knob_caps.stl` onto a slider lever, set `lever_tol` in `case/scad/parts.scad` to the one that grips, then print `knobs-caps/fader_cap.stl` 3 times, upside down. There are no brass inserts: the 10 M3 screws cut their own thread: 8 round posts in the frame (2 under the plate's front corners, 2 under the dock's strip, 4 at the bottom for the lid) and 2 holes in the dock's screw blocks, under the plate's back corners.

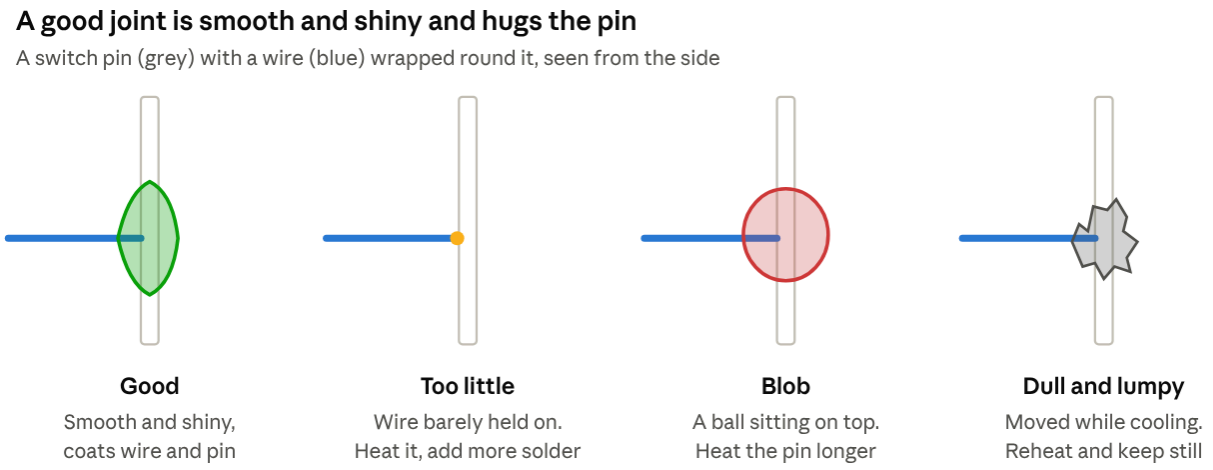
1. While the frame is still empty, drive an M3 x 8 mm screw all the way into each of the 8 posts of the frame and the 2 holes in the dock's screw blocks, then back it out.
2. It should turn firmly without forcing. If it binds hard or a post cracks, raise `pilot_d` in `case/scad/common.scad` (now 2.8) and reprint. If it spins without gripping, lower it.

Step 2: Learn to solder

Practise on 5 scrap wire joints before touching the real parts. It takes about 15 minutes to get the feel.

Stay safe: the iron tip is hot enough to burn instantly. Always put it back in its stand, work with a window open so you don't breathe the smoke, and wash your hands after handling solder.

1. **Heat both parts.** Touch the iron tip so it presses on the pin and the wire at the same time. Count to 2.
2. **Feed the solder into the joint**, not onto the iron. It melts and flows around both parts. About 3 mm of solder is plenty.
3. **Take the solder away first, then the iron.**
4. **Don't move anything for 3 seconds** while it cools.



four kinds of solder joint · only the first is right

A good joint is shiny and smooth. Anything else: put the iron back on for 2 seconds and let it flow again.

Two habits that make it easy:

- Before joining, melt a little solder onto each bare wire end first ("tinning"). Then the joint only needs heat.
- Wipe the iron tip on the damp sponge or brass wool every few joints, so it stays shiny.

Step 3: Fit the switches, knobs, sliders and screens

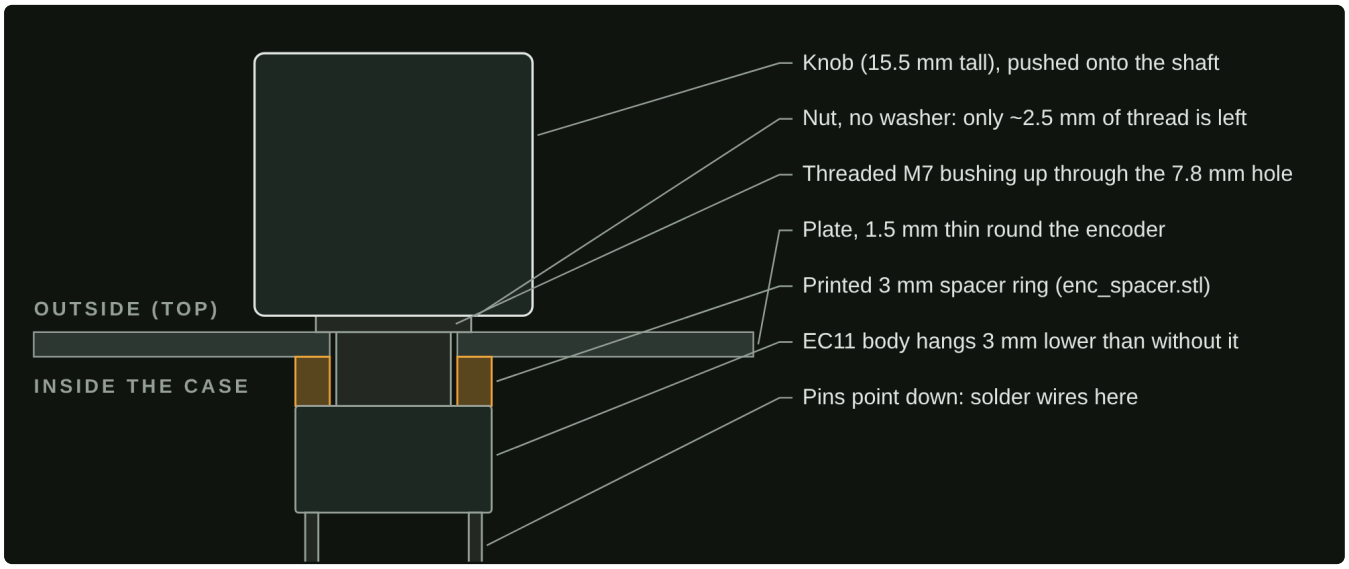
All 22 parts go into the plate before any soldering, because the wires join them together in place. The plate is loose from the frame, so you work on it on the table.

Switches

1. Hold the plate the right way up, key holes facing you, its back edge, where the phone dock goes, away from you.
2. Turn each switch so its **two metal pins point toward the back** (the phone slot). All 12 the same way.
3. Push it straight down into a square hole until it clicks. The clips on its sides hold it.

Knobs (encoders)

The encoder has no clips. A printed 3 mm spacer ring under the plate makes it sit lower, and the nut on top clamps it to the case.



1. Find the small metal tab sticking up next to the threaded part. **Snip it off with flush cutters** or bend it flat with pliers. There is no slot for it, so left alone it tilts the encoder.
2. Take the nut and washer off the encoder.
3. Slide a spacer ring (`enc_spacer.stl`) onto the threaded part. From **below** the plate, push the threaded part up through a round hole, ring against the plate. Turn the encoder so its **row of 3 pins faces the front** (toward its screen) and the 2 pins face the back.
4. On the outside, screw the nut down **without the washer** (only about 2.5 mm of thread pokes out), with your fingers, then a quarter turn more with pliers. Too tight cracks the plastic. If the nut can't catch the thread, hot-glue the ring to the plate and the encoder to the ring instead.
5. Do the same for the second encoder.

Leave the knob caps off for now. They go on in Step 11.

Sliders

1. Turn the plate upside down on a soft cloth.
2. Set each slider into its long pocket with the lever sticking up through the slot. The two small pegs go into the holes at the ends of its metal body. Put the end with pins 1 and 2 toward the back; the single pin 3 then sits at the front, next to the slider's screen.
3. Put a dab of hot glue over each peg.

Screens

1. Still upside down, lay each screen glass-down in its pocket, the row of 4 pins toward the back, so the 4 pegs go through its corner holes.
2. Put a dab of hot glue on each peg tip. Keep glue off the glass.

3. Leave the address resistor on the back as it came (0x3C; the label may say 0x78). All five use the same address; the multiplexer keeps them apart.

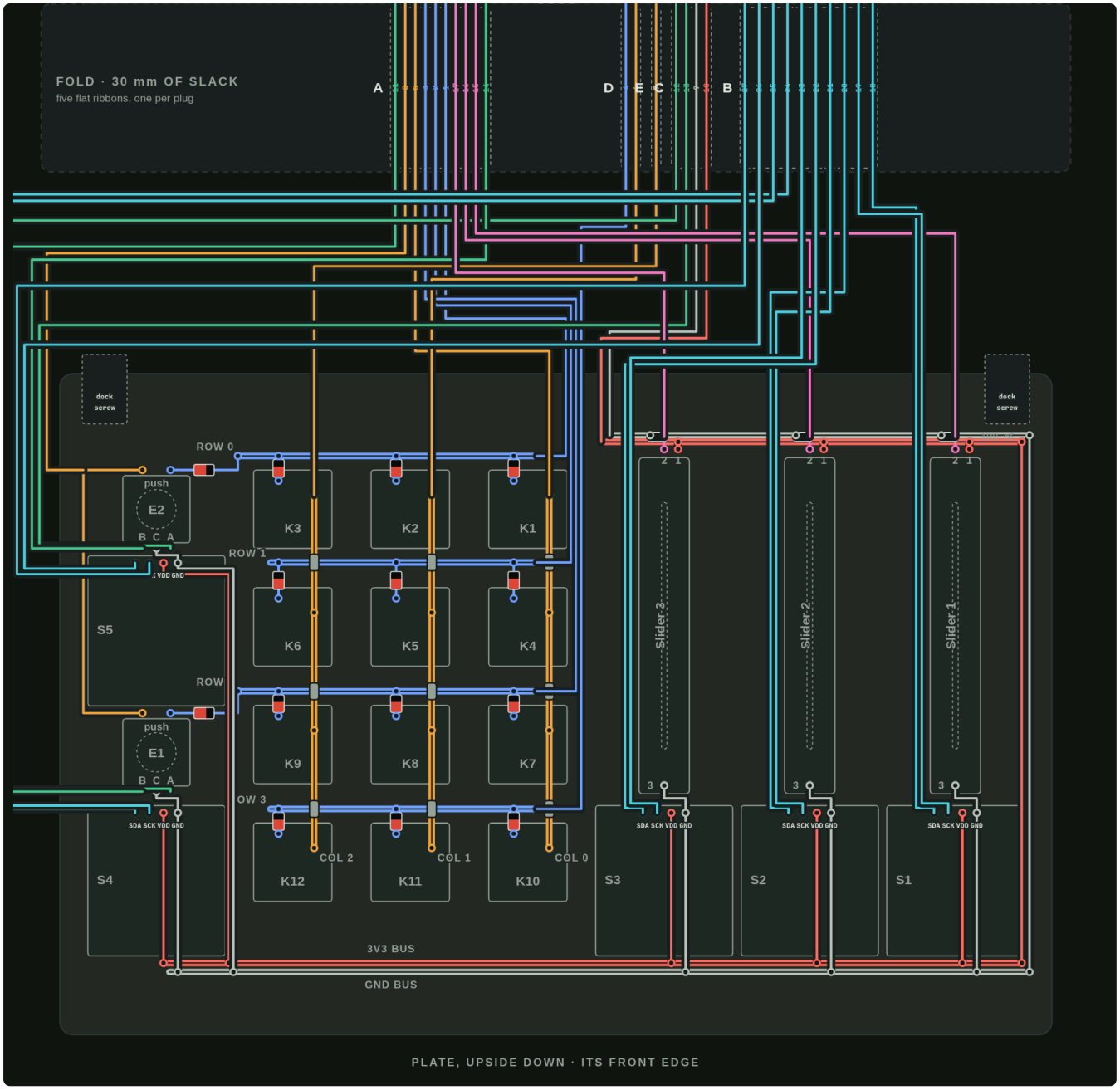
Step 4: Lay the copper buses

Keep the plate upside down on the cloth. From now on you work on the underside, and **everything is mirrored**: K1 is now at the top right of the key block, the sliders are on the right, and the knob column (both knobs and their screens) is at the far left. Keep the back (where the phone dock goes) facing away from you.

The buses are bare solid copper wire that many parts solder onto: one per row of keys, one per column, one **GND** and one **3V3**. Lay them first; every other wire on the plate is a short jumper from a part to a bus. Wherever a row crosses a column, **slide a short piece of sleeve or heat-shrink over the column first**: bare copper touching bare copper is a short.

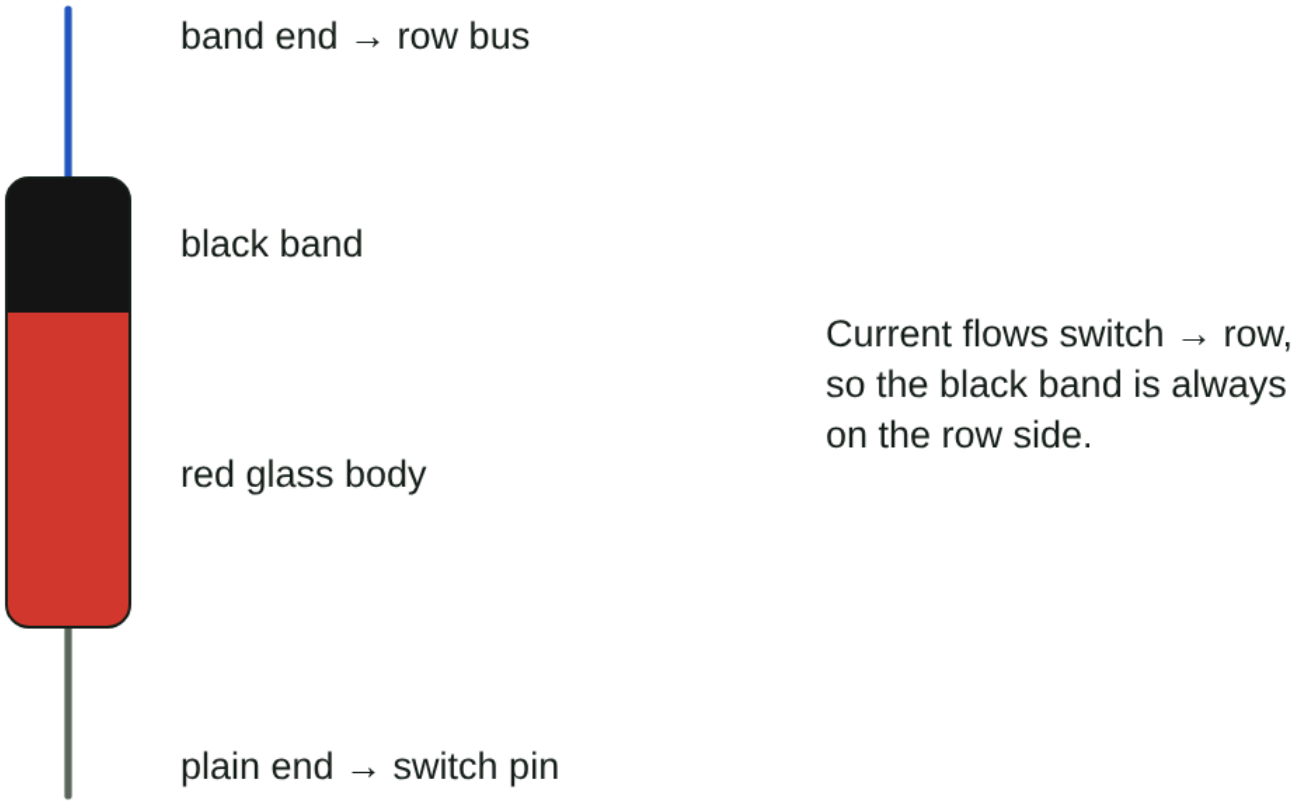
1. **Columns 0 to 2.** Cut about 12 cm of copper per key column. Slide on 3 pieces of sleeve, about 8 mm each. Lay it along the column so it touches the switch pin nearer the front of each key (K1, K4, K7, K10 for column 0), then solder it to all 4 pins. Leave the back end long: its numbered wire leaves from there (Step 8).
2. **Rows 0 to 3.** One piece across each key row, a few mm behind the switches, resting on the column sleeves. Rows 0 and 2 run on into the gap beside the knob column, to E2′s and E1′s push pins; rows 1 and 3 stop at the keys. Tack each row down with a dab of hot glue every few cm; the diodes hold it for good in Step 5.
3. **Column 3.** Not a bare bus: an insulated wire along the knob-column wall, from E1′s push pins to E2′s and on to the back edge. It goes on in Step 5.
4. **GND and 3V3.** Two pairs of wires, a few mm apart, GND always the outer one: one pair across the front, just in front of the screens′ pin rows, the other across the back ends of the sliders. Join the pairs with two insulated wires up the slider-side wall, GND to GND and 3V3 to 3V3. Bare copper never runs over a screen or any other board. Tack them down with hot glue.

The drawing below is the plate when Steps 4 to 7 are done: thick lines are the buses, thin lines the jumpers, grey blocks the sleeves. The numbered wires at the top run on to the lid in Step 8. The same drawing, clickable, is on the wiring page: pick any part to see only its wires.

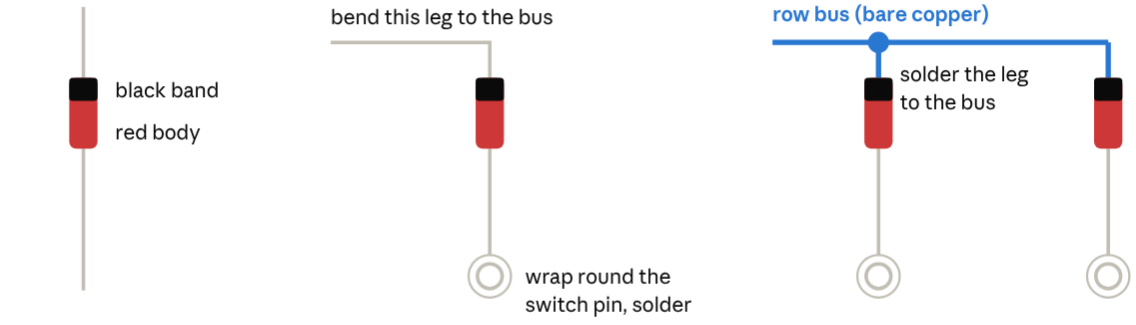


Step 5: Diodes and column joints

Every key gets one diode, from its free switch pin to its row bus. The rule never changes:



Black band up: that leg goes to the row bus



1. Band end up
plain end down

2. Bend and solder
one diode per key

3. Solder to the row bus
snip the spare leg

preparing a diode · 3 moves per key

1. Start with K1 (top right). Wrap the diode's **plain end** around the switch pin that is closer to the back (the one not on the column bus) and solder it. The black band points up, toward the row bus.
2. Bend the band-end leg so it lies against the row 0 bus and solder it there. Snip the spare leg.
3. Do the same for K2 and K3, then K4 to K6 on **row 1**, K7 to K9 on **row 2** and K10 to K12 on **row 3**.

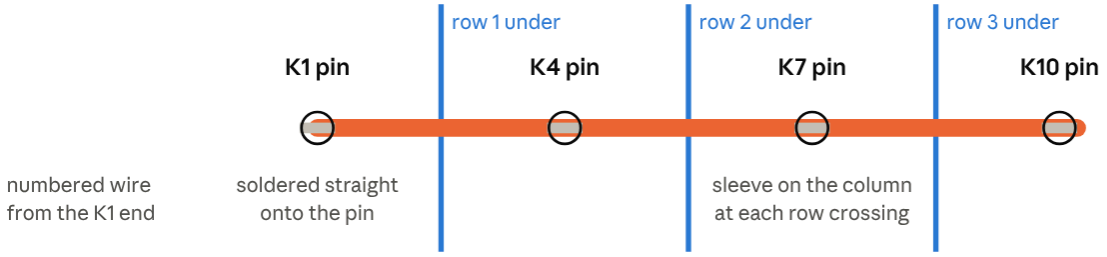
4. The knob pushes join a row too: knob 2 (E2, at the back of the knob column) on **row 0**, knob 1 (E1, lower down) on **row 2**. On each knob, use the push pin **nearer the keys** (the right one of the 2-pin side, as you look at the underside). Its diode stands upright like the others, band end to the row bus.
5. Once every diode is in, check each row bus still sits on its sleeves at every column crossing.

Column joints.

The other pin of each switch is already soldered to its column bus (Step 4). Here is what one column looks like, and which pins each one joins:

The column bus is bare copper, with a sleeve where each row crosses

Column 0, laid out straight: from the K1 pin at the back to the K10 pin at the front



Sleeves:

slide 3 pieces of sleeve or heat-shrink, about 8 mm each, onto the column before you solder it.

Move each one under the spot where a row will cross; shrink heat-shrink gently with the side of the iron.

one column bus · 4 joints, 3 sleeved crossings

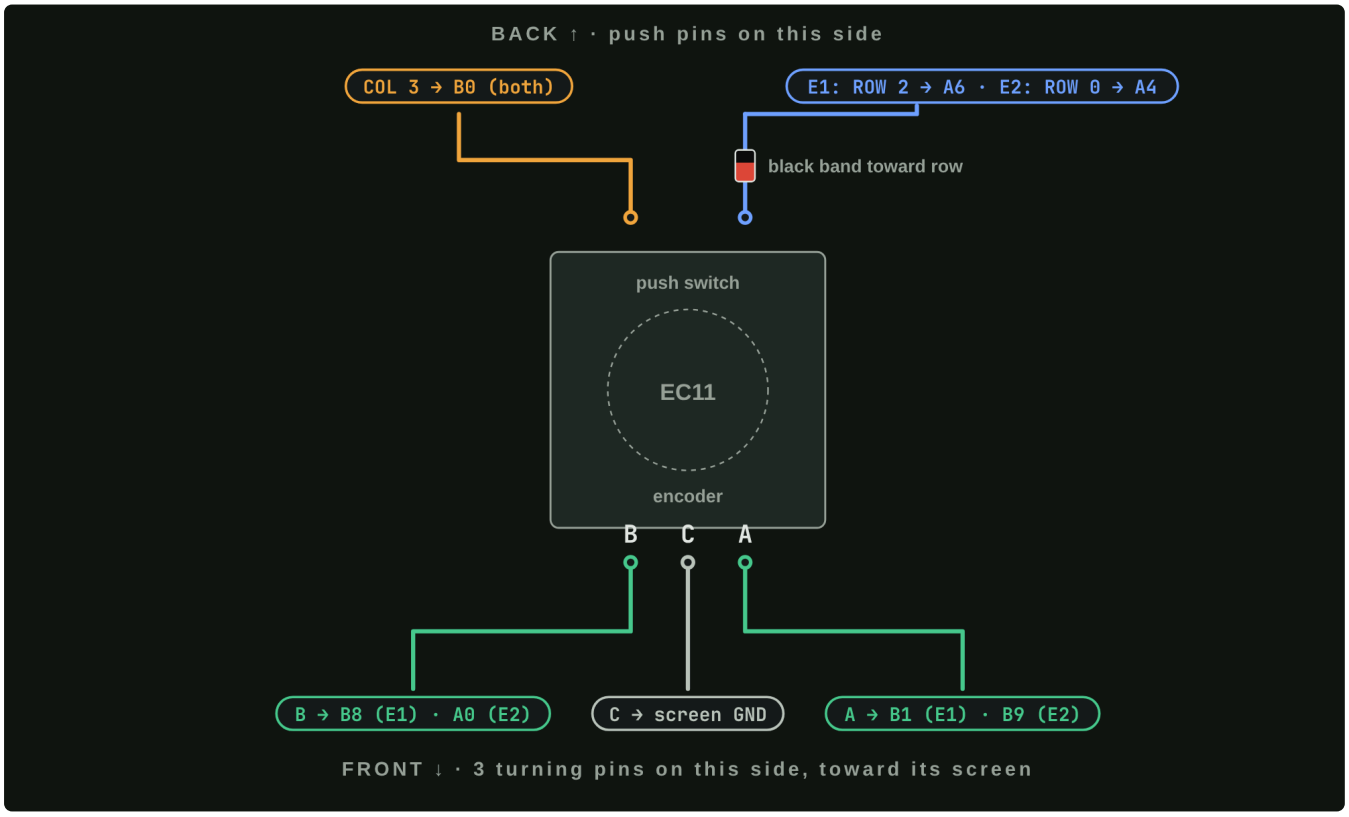
Column	Joins these pins (from the back)	Goes to BlackPill pin
Column 0	K1, K4, K7, K10	A7
Column 1	K2, K5, K8, K11	A8
Column 2	K3, K6, K9, K12	A15
Column 3	E2 push, E1 push (the knob column)	B0

1. Check each column bus is soldered to all 4 of its switch pins.
2. Column 3 joins only the two knob pushes, through an insulated wire: solder it to E1's **left** push pin (the one without a diode), run it along the knob-column wall to E2's left push pin, strip a short window there and solder it, then carry on to the back edge as numbered wire **8**.
3. Check every place a row crosses a column: a sleeve must sit between them. If bare metal touches, those keys will act wrong.

The rows and columns reach the BlackPill through numbered wires 1 to 8, in Step 8.

Step 6: The knob pins

Each knob has 5 pins. The 2 push-button pins on the back side are done (Step 5). Now the 3 on the front side, which tell the BlackPill which way the knob turns: C goes to the GND pin of the knob's own screen, A and B are numbered wires to the BlackPill.



Seen from below, the 3 front pins read **B, C, A** from left to right. C is always the middle one.

Pin	Knob 1 (E1, front of the knob column)	Knob 2 (E2, back of the knob column)
A	wire **11** to B1	wire **13** to B9
C (middle)	short jumper to S4's GND pin	short jumper to S5's GND pin
B	wire **12** to B8	wire **14** to A0

1. C: a short jumper forward to the GND pin of the knob's own screen, just in front of it (S4 for E1, S5 for E2).
2. A and B: a soft silicone wire on each, long enough to reach the lid with 3 cm to spare. Run them out to the knob-column wall and up it to the back edge.
3. Tape each loose end with its number (11 to 14, see the table). They plug onto the BlackPill in Step 8.

No diodes on these pins. If a knob later turns the wrong way, don't re-solder: it is a one-line change in the software.

Step 7: Wire the sliders and the screens

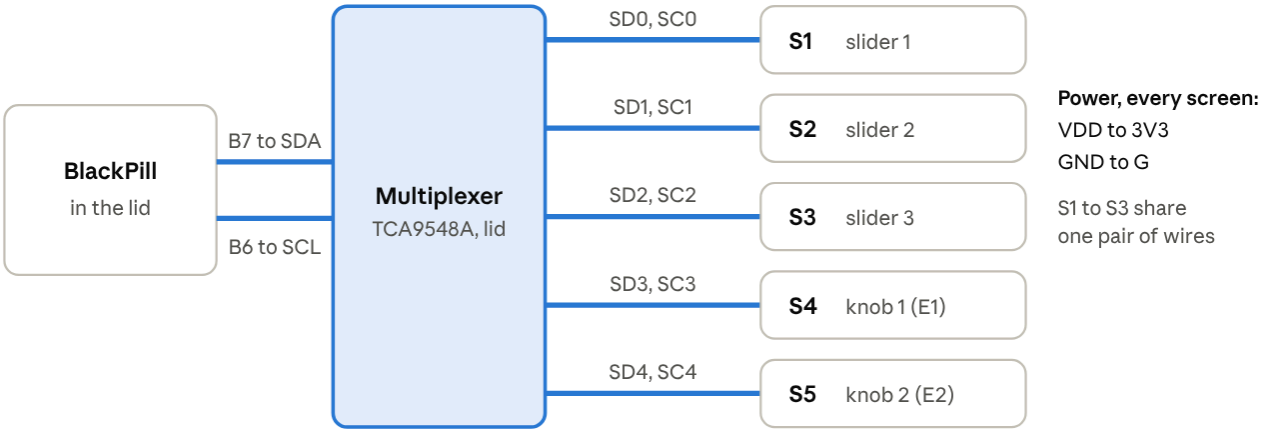
Still on the upside-down plate. Power pins take short jumpers to the buses; each signal pin takes a numbered wire with a tape label, which plugs in at the lid in Step 8.

Sliders. Each slider has 3 pins, numbered on its metal body: pins 1 and 2 at the back end, pin 3 alone at the front end. Pins 1 and 3 are the two ends of the track; pin 2 is the middle one that moves.

1. Pin 1: a short jumper to the **3V3** bus running across the slider ends.
2. Pin 3, at the front end: a short jumper to the **GND** pin of the slider's own screen, just in front of it.
3. Pin 2: a numbered wire straight back to the back edge: **15** for slider 1 (the one on the right from below), **16** and **17**.
4. Solder a 100 nF capacitor from pin 2 to the **GND** bus across the slider ends, right beside pin 2 (pin 3 is the same GND, but at the far end). It has no + or −, so either way round.

Screens. Each screen has 4 pins: GND, VDD, SCK, SDA; check the print on yours. All 5 answer to the same address, so each one's SCK and SDA go to its own channel on the multiplexer, a small board that sits in the lid next to the BlackPill (Step 8). Only the power is shared.

Each screen gets its own channel on the multiplexer



Each blue line is 2 wires: the screen's SDA to SDn and its SCK to SCn. The mux also takes 3V3 and G.

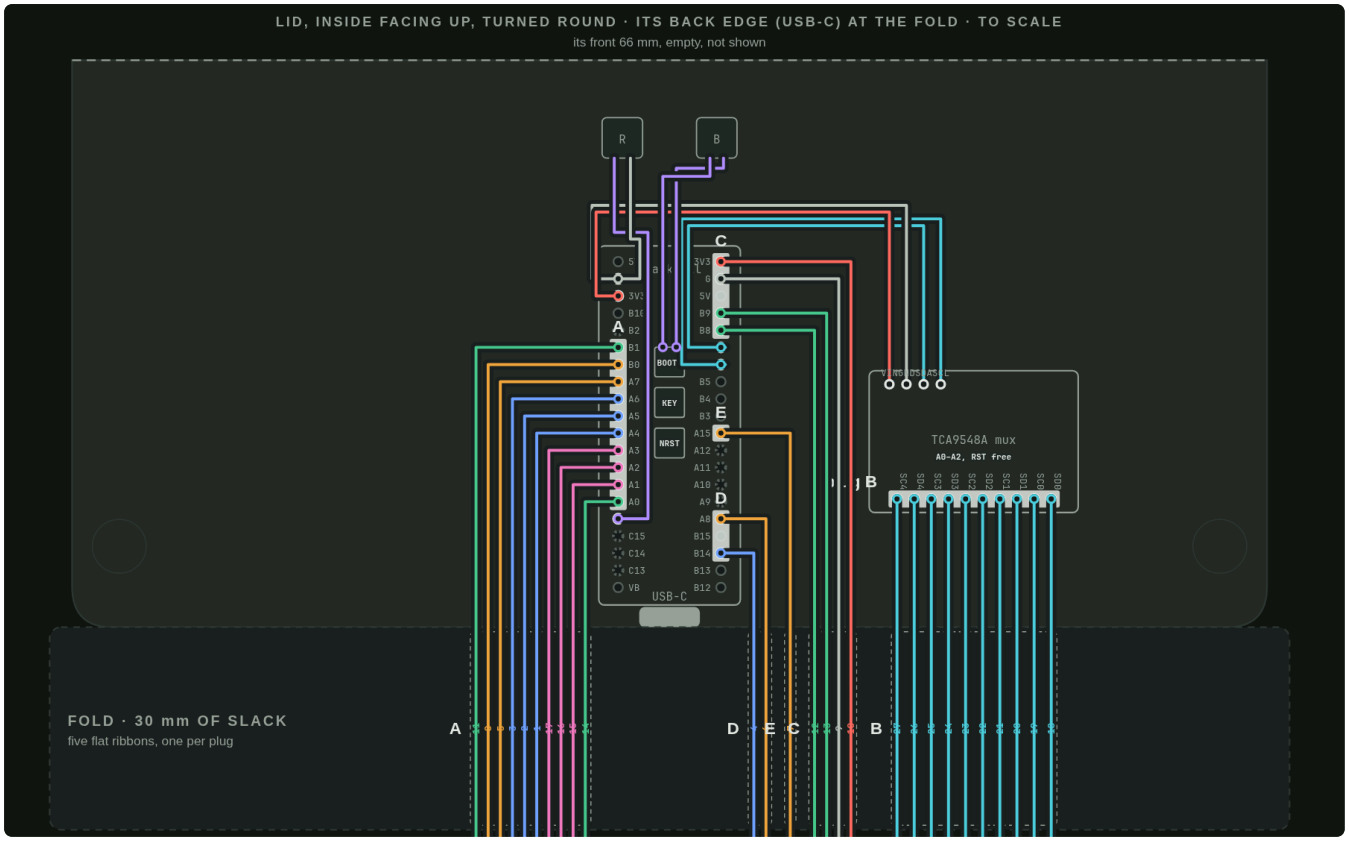
screen wiring · one multiplexer channel per screen

Screen pin	Goes to
GND	the GND bus, by a short jumper
VDD	the 3V3 bus, by a short jumper
SDA	a numbered wire to its multiplexer pad: S1 **18** to SD0, S2 **20** to SD1, S3 **22** to SD2, S4 **24** to SD3, S5 **26** to SD4
SCK	a numbered wire to the matching pad: S1 **19** to SC0, S2 **21**, S3 **23**, S4 **25**, S5 **27** to SC4

1. VDD and GND: short jumpers down to the 3V3 and GND buses just in front of the pin rows. S5 sits between the knobs, far from the front pair: its two run down the gap between the keys and the knob column to the front pair.
2. SDA and SCK: numbered wires to the back edge. S1 to S3 step back and run up the gap beside their slider; S4 and S5 run out to the knob-column wall and up it.
3. Power the screens from **3V3** only, never 5V.

Step 8: The lid and the harness

Now the 27 numbered wires go to the lid. They are not soldered there: they gather into **five flat ribbons** that end in multi-pin Dupont plugs, **A** to **E**, which push onto header pins on the BlackPill and the multiplexer. The lid unplugs, and the wiring stays tidy. The pin names are printed next to each hole: **A4** means pin PA4, **B14** means PB14, **G** is ground, **3V3** is the 3.3 V supply, **R** is reset.



Lay the lid above the plate as in the drawings: inside up, turned round, its back edge (where the USB-C plug sits) against the plate's back edge. That edge is the **fold**: the case opens and closes about it like a book, and the five ribbons cross it. The drawings are to scale, so you can hold a ribbon against them to check its length. Each wire goes from the plate end in this table to its pin, in its plug:

Wire	On the plate	In the lid	Plug
1	Row 0 bus, its end by column 0	A4	A
2	Row 1 bus, its end by column 0	A5	A
3	Row 2 bus, its end by column 0	A6	A
4	Row 3 bus, its end by column 0	B14	D
5	Column 0 bus, its back end	A7	A
6	Column 1 bus, its back end	A8	D
7	Column 2 bus, its back end	A15	E
8	Column 3 wire (knob pushes), at the wall by E2	B0	A
9	GND back bus, its end by slider 3	G (left-hand header)	C
10	3V3 back bus, its end by slider 3	3V3 (left-hand header)	C

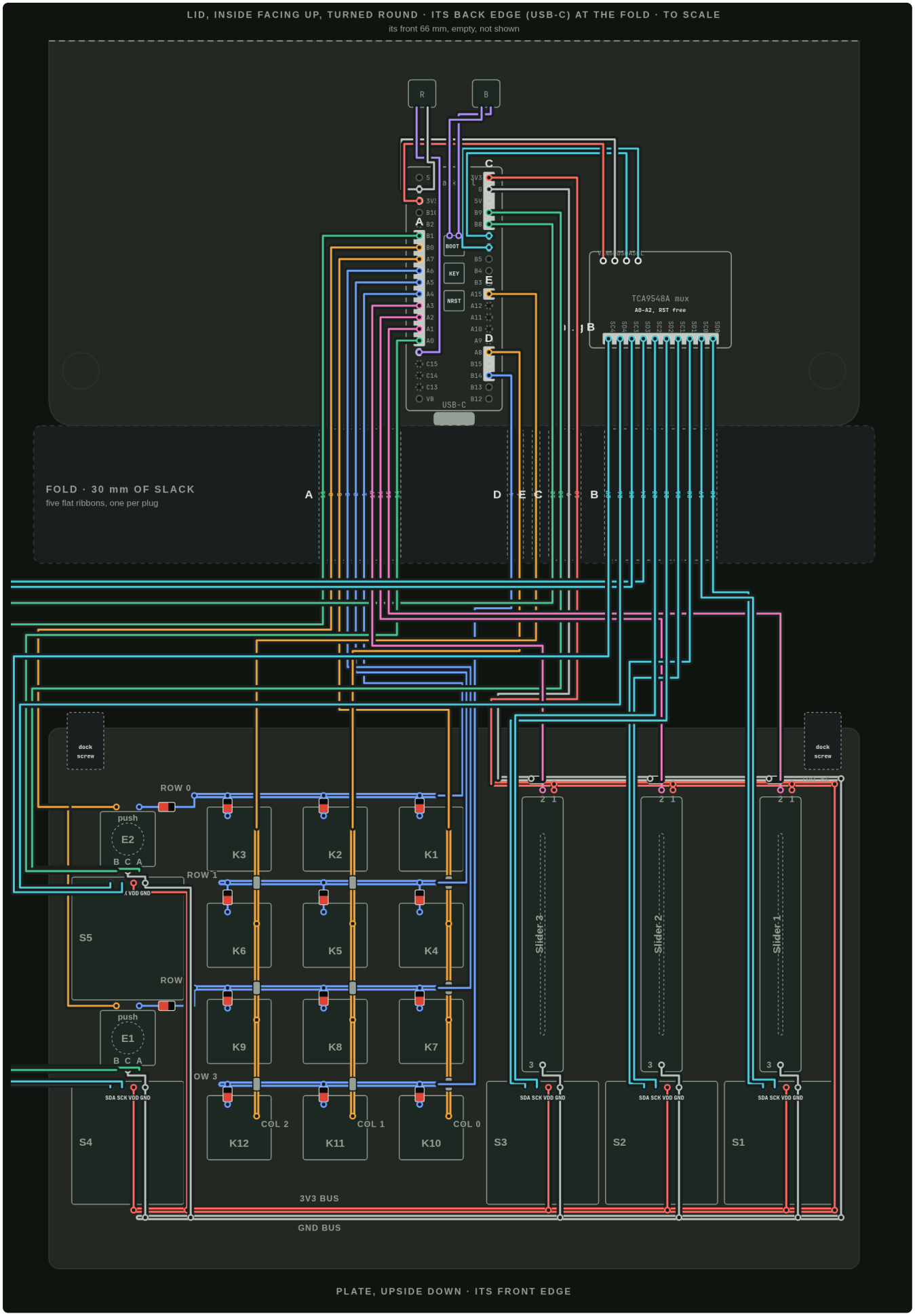
Wire	On the plate	In the lid	Plug
11	E1 pin A	B1	A
12	E1 pin B	B8	C
13	E2 pin A	B9	C
14	E2 pin B	A0	A
15	Slider 1 pin 2	A1	A
16	Slider 2 pin 2	A2	A
17	Slider 3 pin 2	A3	A
18	S1 SDA	mux SD0	B
19	S1 SCK	mux SC0	B
20	S2 SDA	mux SD1	B
21	S2 SCK	mux SC1	B
22	S3 SDA	mux SD2	B
23	S3 SCK	mux SC2	B
24	S4 SDA	mux SD3	B
25	S4 SCK	mux SC3	B
26	S5 SDA	mux SD4	B
27	S5 SCK	mux SC4	B
L1	(lid) mux SCL	B6	–
L2	(lid) mux SDA	B7	–
L3	(lid) mux GND	G (right-hand header)	–
L4	(lid) mux VIN	3V3 (right-hand header)	–
L5	(lid) B button, both legs	the board's own BOOT0 button, two legs	–
L6	(lid) R button, both legs	R and G (right-hand header)	–

The same table is on the wiring page, with a box to tick per wire; hover a row there to see that wire in the drawing.

The five plugs. Each is one Dupont housing, its wires in the order of the header pins under it. Leave the empty slots empty: they keep each plug in its place.

Plug	Header pins, in order	Wires
A, 10 pins	BlackPill right-hand header: A0, A1, A2, A3, A4, A5, A6, A7, B0, B1	14, 15, 16, 17, 1, 2, 3, 5, 8, 11
B, 10 pins	multiplexer: SD0, SC0, SD1, SC1, SD2, SC2, SD3, SC3, SD4, SC4	18 to 27
C, 5 pins	left-hand header: B8, B9, (5V, empty), G, 3V3	12, 13, –, 9, 10
D, 3 pins	left-hand header: B14, (B15, empty), A8	4, –, 6
E, 1 pin	left-hand header: A15	7

Every wire at once. The whole case laid out for wiring, to scale: lid on top, plate below, the fold between. The wires gather along the plate's back edge into five ribbons, cross the fold and end in plugs A to E; each wire's number is printed on its line in the fold band. On the wiring page, click a number, part or bus to see its whole run.


LID, INSIDE FACING UP, TURNED ROUND · ITS BACK EDGE (USB-C) AT THE FOLD · TO SCALE
its front 66 mm, empty, not shown
R
B
C
A
E
D
BOOT
KEY
NRST
USB-C
TCA9548A mux
A0–A2, RST free
FOLD · 30 mm OF SLACK
five flat ribbons, one per plug
A
D
E
C
B
dock screw
dock screw
ROW 0
ROW 1
ROW
OW 3
push
E2
E1
B C A
VDD GND
SDA SCK VDD GND
K3
K2
K1
K6
K5
K4
K9
K8
K7
K12
K11
K10
COL 2
COL 1
COL 0
S5
S4
S3
S2
S1
Slider 3
Slider 2
Slider 1
3V3 BUS
GND BUS
PLATE, UPSIDE DOWN · ITS FRONT EDGE

1. **Header pins.** Solder male header strips into both rows of the BlackPill, long pins up, the short ends through the board, and snip what sticks out underneath. Do the same on the multiplexer: one strip on its channel pads (SD0 to SC4), one on VIN, GND, SDA and SCL.
2. **Lid wires** L1 to L6: the multiplexer and the buttons (below).
3. **One ribbon at a time.** Take the wires of one plug from the table, lay them along the plate's back edge as in the drawing, side by side in the plug's order, and tape or lace them into a flat ribbon. With rainbow ribbon cable, peel each wire off where it leaves for its part.
4. Hold the ribbon across the fold to its pins with about 3 cm of slack, cut it, crimp a female Dupont pin onto each wire and push them into the housing in the table's order.
5. Put a dot of paint on pin 1 of each housing and of the header under it, then push the plug on.
6. **Leave these pins free:** A9 and A10 (they upset USB loading), A11 and A12 (USB), B2, C13, C14 and C15. A0 also has the board's own KEY button on it, which is fine.

Multiplexer. Hot-glue the TCA9548A board in the lid on the left of the BlackPill (the side with B6 and B7), pads facing up and its channel pads toward the back wall, as in the drawing. Its pads are printed on it; follow the names. Check its channel pads run SD0, SC0, SD1, SC1 and so on in that order, so plug B fits; if they don't, use single plugs there.

1. SCL to B6 and SDA to B7, short wires up the inside of the BlackPill.
2. VIN to 3V3 and GND to G, both on the right-hand header, round the BlackPill's front end.
3. The screens' SDA and SCK wires arrive in plug B: S1 on SD0 and SC0, up to S5 on SD4 and SC4.
4. Leave A0, A1, A2 and RST unconnected. The board then answers at address 0x70, which the firmware expects.

Buttons. The two 6 x 6 mm buttons go in the lid, upside down, and you press them from below with a pen tip. Each button has 4 legs that work in pairs: use two legs that are **diagonally opposite**, so it works whichever way it is turned.

- **R (reset):** one leg to **R**, the other to **G**, both on the right-hand header.
- **B (BOOT0):** BOOT0 has no pin hole, so B is wired across the board's own small BOOT0 button. Set the meter to beep, find two legs of that button that beep only while it is pressed, and solder a thin wire to each. Their other ends go to the B button.

To load firmware: hold B, tap R, let go of B.

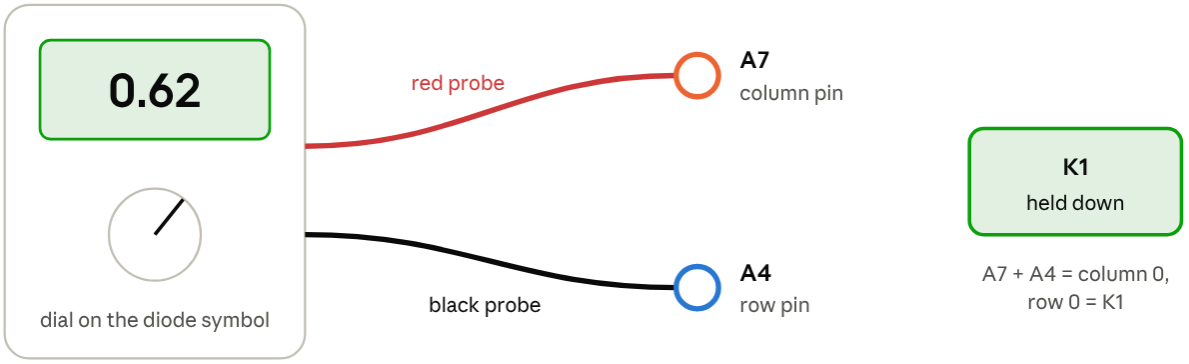
Step 9: Test everything with a multimeter

Testing now takes 15 minutes and finds almost every mistake while it is still easy to reach. Keep the USB cable unplugged.

First, a safety check: set the meter to the beep (continuity) setting and touch the probes to **3V3** and **G**. It must **not** beep. If it beeps, something is shorted: look for a stray blob of solder or a bare wire touching before going further.

Keys. Turn the dial to the diode symbol (a triangle with a line). For each key: red probe on its column pin, black probe on its row pin, then press and release the key.

Key pressed: about 0.6 on the meter. Key released: nothing.



testing one key · meter in diode mode

	A7 (column 0)	A8 (column 1)	A15 (column 2)	B0 (column 3)
A4 (row 0)	K1	K2	K3	Knob 2 push
A5 (row 1)	K4	K5	K6	–
A6 (row 2)	K7	K8	K9	Knob 1 push
B14 (row 3)	K10	K11	K12	–

A number while the key is **released** means that row and column are touching somewhere. **Nothing** while pressed usually means a diode is the wrong way round or a joint is loose.

- [ ] Buses: meter on beep, one probe on each row's BlackPill pin (A4, A5, A6, B14) and the other on each column's (A7, A8, A15, B0), no key pressed: none may beep. A beep means a sleeve has slipped at that crossing
- [ ] All 12 keys and both knob pushes read about 0.6 pressed, nothing released
- [ ] Knobs: meter on beep, probes on **B1** and **G**, turn knob 1 slowly: it clicks on and off. Same for B8, then B9 and A0 for knob 2

- [ ] Sliders: meter on resistance, probes on **A1** and **G**. Moving slider 1 changes the reading smoothly from 0 to about 10 kΩ. Same for A2 and A3
- [ ] Screens: meter on beep. Each screen's VDD beeps with 3V3 and its GND with G; its SDA and SCK beep only with their own multiplexer pads (S1 with SD0 and SC0, and so on). The multiplexer's SDA beeps with B7 and its SCL with B6

Step 10: The firmware

The firmware (QMK for the BlackPill) is **not written yet**; this step gets the file and exact commands once it is. Loading it will work like this:

1. Plug the BlackPill into your computer with a USB-C cable that carries data. The case can still be open.
2. Hold the **B** button, tap **R**, let go of B. The board waits to be loaded (on Linux, `lsusb` shows `STM Device in DFU Mode`).
3. Load the file with the command given here. The board restarts by itself.
4. Press a few keys with a text editor open.

If the board won't wait to be loaded, try a few more times. As a last resort the ST-Link programmer loads it through the 4-pin header at the end of the board.

Step 11: Close the case

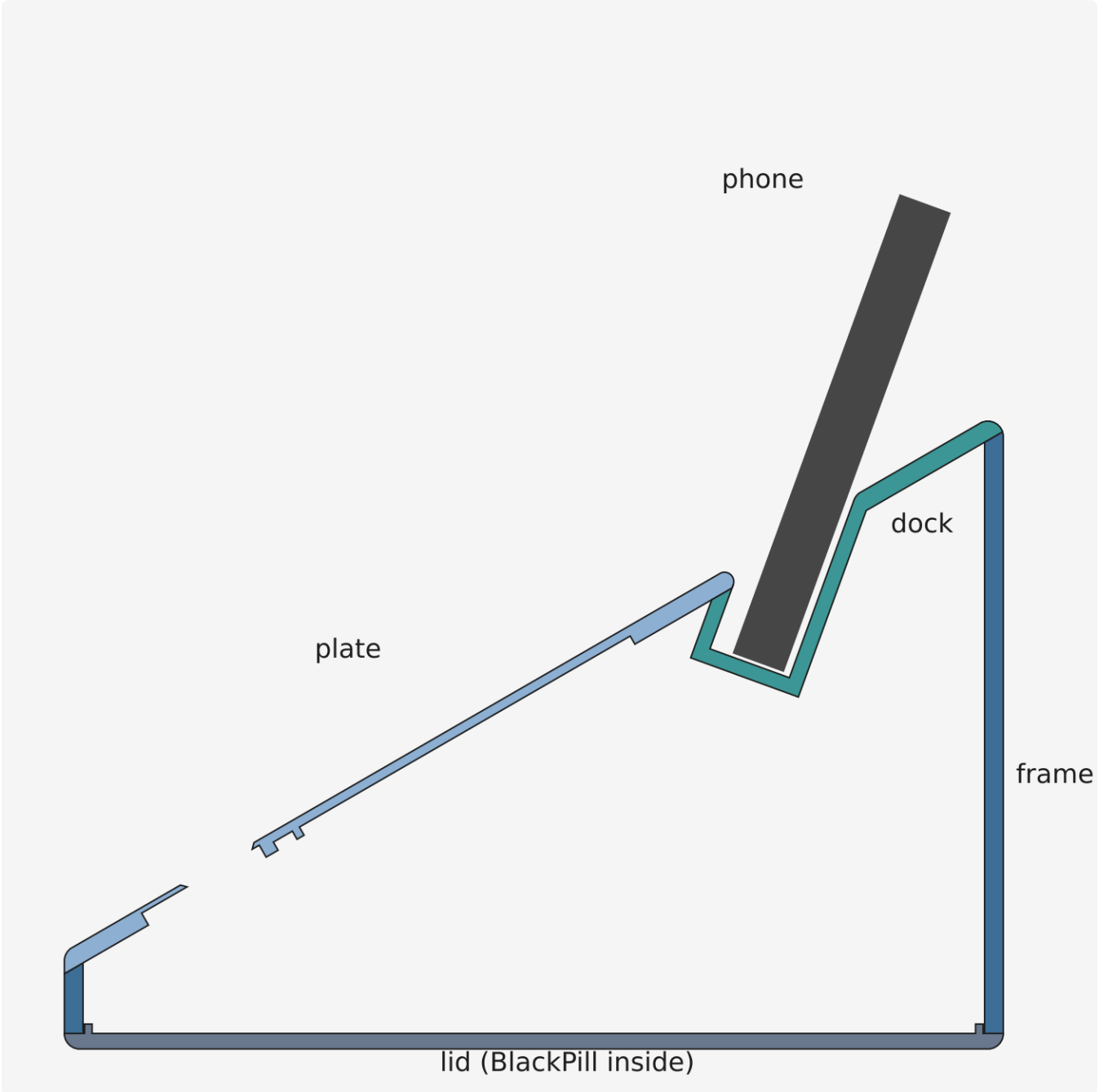
Only close the case once everything passed Step 9 and the firmware works.

1. Set the BlackPill into the cradle in the lid, chips up, USB-C plug lined up with the hole in the back wall. A dab of hot glue holds it. The multiplexer is already glued beside it (Step 8).
2. Hot-glue the B and R buttons upside down into their pockets in the lid (marked **B** and **R** underneath), the button part poking down through the small hole.
3. Fold the wires in gently and slide the phone dock down into the notches in the frame's side walls, then set the plate onto the frame in front of it. Nothing should be pinched.
4. Screw the dock's strip down with 2 M3 x 8 mm screws, then the plate with 4: 2 at its front corners into the frame, 2 at its back corners into the dock's screw blocks. Turn each one backwards until it clicks into the thread it cut in Step 1, then screw it in until snug. Too tight strips the plastic.
5. Screw the lid on from below with the other 4 screws, the same way.
6. Stick on the rubber feet. Push the keycaps onto the switches, the knobs onto the encoder shafts and the caps onto the slider levers.

Using it: the phone slot and the sliders

Phone slot. A slot across the back of the deck, behind the keys. Your phone sinks 15 mm in and leans back at 70°. It is sized for an iPhone 16 Plus in a case up to 11 mm thick; for a thicker case raise `phone_t` in `case/scad/deck.scad`.

- **Landscape:** the slot is open at both sides, so the phone, which is wider than the deck, sticks out about 7 mm each side. The charging port is at the side, so you can charge it in the slot.
- **Portrait:** the phone stands in the middle. Its charging port is below the slot floor, so it can't charge while standing in portrait.



Set up the computer (Linux). The computer sees the sliders as a game controller with 3 axes. A small helper program turns them into volume controls:

1. Open a terminal in `macropad/helper`.
2. Run `./macropad_sliders.py --list`. A line ending in "<- sliders" means the computer sees them.
3. Run `./macropad_sliders.py --apps` while music plays, to see the names of programs making sound.
4. Open `sliders.toml` and set what each slider does. Out of the box: slider 1 = music (Spotify, Firefox), slider 2 = voice chat (Discord, Zoom), slider 3 = overall volume.
5. Run `./install.sh`. The helper now starts by itself every time you log in.

The sliders are new and not yet tested on real hardware. If something is off, the table below and the README's firmware section are the place to start.

If something doesn't work

Find the line that matches what you see. Most problems are one loose joint or one diode the wrong way round.

What you see	Most likely cause	What to do
One key does nothing	Diode backwards, or a loose joint on that key	Test it as in Step 9. Check the black band faces the row. Reheat both of that key's joints.
A whole row of keys does nothing	Row wire not reaching its pin, or a broken leg joint along the row	Reheat the joint at the BlackPill (A4, A5, A6 or B14), then each diode's joint on the row bus.
A whole column does nothing	Column wire not reaching its pin	Reheat the joint at A7, A8, A15 or B0.
One key types what another should	Two wires swapped at the BlackPill	Compare your pins with the table in Step 8.
Pressing one key types two things	A row bus and a column bus touching where they cross	Find the crossing where the sleeve has slipped and slide it back, or add a piece of heat-shrink.
Knob turns the wrong way	A and B swapped	No soldering: swap the two actions on that knob in the firmware.
Knob skips or jumps	Loose wire on A, B or C	Reheat the 3 front pins and the ground wire.

What you see	Most likely cause	What to do
Slider works upside down	Pins 1 and 3 swapped	No soldering: set `invert = true` in `sliders.toml` and restart the helper.
Slider jumps or flickers	Loose joint on pin 2 or G, or its 100 nF capacitor missing	Reheat the joints; check the capacitor sits between pin 2 and the GND bus, and pin 3's jumper reaches its screen's GND pin.
Sliders do nothing on the computer	Firmware not loaded, or the helper is not running	Run `./macropad_sliders.py --list`; then `journalctl --user -u macropad-sliders -f`.
A program's slider does nothing	Name doesn't match	Check the name with `--apps` while it plays sound and put that word in `apps`.
No screen lights up	The multiplexer has no power, or its SDA and SCL are swapped	Check VIN to 3V3, GND to G, SDA to B7, SCL to B6, and that A0 to A2 are left unconnected.
One screen stays dark	Loose SDA or SCK wire, or wired to the wrong multiplexer pads	Check its 4 wires, and that its SDA and SCK sit on the same channel number (SD2 and SC2, say).
Screen doesn't sit flat	Glass pocket or pegs don't match your screen	Measure it and change the `scr_` sizes in `case/scad/common.scad`; test with `fit_screen.stl`.
Board won't wait to be loaded	B pressed too briefly, or a charge-only USB cable	Hold B the whole time you tap R. Try another cable. Last resort: the ST-Link.
Switch falls out or won't click in	Hole size off	Adjust `switch_cut` (Step 1) and reprint the plate.